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SENEGAL

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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION AND EIGHTH REVIEW UNDER THE POLICY SUPPORT INSTRUMENT—DEBT SUSTAINABILITY ANALYSIS

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Senegal remains at a low risk of debt distress. Under the baseline scenario, which is consistent with higher program ceilings for non-concessional and semi-concessional borrowing, all the debt burden indicators remain below their policy-dependent indicative thresholds, and debt ratios in present value terms are lower than in the previous debt sustainability analysis (DSA). Policy-dependent thresholds were increased as Senegal was reclassified to 'strong' performer based on higher average CPIA score in 2011-2013.[1] *The probability approach also shows a more favorable outlook. The stress tests result in two spikes in debt service to revenue ratio, corresponding to the repayment of two Eurobonds, which lead to a small and temporary breach of the threshold. The DSA, however, suggests that there is not much space for higher fiscal deficits, if the low risk rating is to be preserved. It also indicates a need for caution in the recourse to non-concessional borrowing.*[2]

[1] Senegal's CPIA score was 3.825 in 2013, and on average 3.81 over 2011–13. Under the debt sustainability framework rules, this corresponds to a "strong" performance.

[2] The DSA presented in this document is based on the LIC DSF Guidance Note (2013).

BORROWING PLAN AND UNDERLYING ASSUMPTIONS

1. The authorities have continued to strengthen their capacity to manage debt and assess project loans. Senegal recently recorded improvement in its sub-score on debt management under the Country Policy and Institutional Assessment (CPIA). A new public debt directorate has been created, combining units that previously managed domestic debt and external debt separately. In addition, Senegal's first medium-term debt strategy was completed in the fall of 2012. The strategy essentially aims at reducing rollover risks by extending the maturity of debt issued on the regional market—which has a very short average maturity—as well as at giving priority to concessional financing to keep borrowing costs low. An updated medium-term second debt management strategy is being finalized along the same lines. Progress is underway to improve project appraisal and selection, in particular by developing capacity to conduct cost-benefit analysis. In light of these favorable developments, Senegal was upgraded to the "strong capacity" category during the 2013 assessment of macroeconomic and public financial management capacity (see Classification of Low-Income Countries for the Purpose of Debt Limits in Fund-Supported Programs).

Table 1. Total External Debt, Central Government
(Percent of Total, as of end of year)

	(Percent of Total, as of end of year)					
	2008	**2009**	**2010**	**2011**	**2012**	**2013**
Total	100.0	100.0	100.0	100.0	100.0	100.0
Multilateral creditors	61.2	65.1	66.4	60.2	62.1	63.7
IDA/IBRD	35.6	29.7	29.2	32.2	30.2	29.6
AfDB/AfDF	7.6	8.3	9.0	10.5	9.8	11.0
IMF	2.6	6.8	10.8	0.0	6.1	7.4
OFID/BADEA/IsDB	8.6	11.9	10.1	10.2	8.4	8.0
EIB	0.9	1.0	1.0	0.9	0.8	0.8
Others	5.9	7.4	6.4	6.4	6.9	6.8
Bilateral creditors	38.7	29.3	27.8	26.3	26.5	26.2
OECD countries	11.2	10.2	9.7	7.7	10.4	10.1
Arab countries	21.2	14.1	10.3	8.1	6.9	6.1
Others	6.3	5.0	7.8	8.4	9.1	8.8
Commercial creditors	0.1	5.6	5.8	13.5	11.4	10.1
Memorandum Item						
Nominal GDP (CFAF billions)	5994	6050	6395	6775	7165	7308

Sources: Senegalese authorities and IMF staff estimates.

2. Senegal has also now been reclassified as a strong performer in the CPIA index. This reclassification has led to an increase of indicative thresholds for each debt burden indicators, which has improved Senegal's debt outlook. The present value of the external debt-to-GDP ratio has been raised to 50 percent from 40 percent in the previous DSA (without remittances), while the ratios for external debt-to-exports and for external debt-to- revenue are 200 and 300 respectively, up from 150 and 250 in the previous DSA. The ratio for PPG external debt service in relation to exports is now 25 percent and 22 percent in relation to revenue compared to 20 percent thresholds for both indicators in the previous DSA (Table 2).

Table 2. PPG External Debt Thresholds

Quality of Policies and Institutions (CPIA)	PV of PPG External Debt in percent of			PPG External Debt Service in percent of	
	GDP	Exports	Revenue	Exports	Revenue
Weak	30	100	200	15	18
Medium	40	150	250	20	20
Strong	50	200	300	25	22

Source: IMF

3. This DSA is consistent with the macroeconomic framework outlined in the staff report. As in the April 2014 DSA, the baseline scenario assumes the implementation of sound macroeconomic and structural policies, leading to an increase in economic growth and a narrowing of fiscal deficits over the long term. Other notable features include:

- **Real GDP growth** is expected to increase to above 5 percent only in 2016 and to accelerate and 7.3 percent on average in 2020-34, from a high base of 7 percent in the medium-term, compared to 6.2 percent medium-term growth in the previous DSA. This assumes efficiency gains from reform implementation under the authorities Plan Senegal Emergent (PSE),[3] which would kick start total factor productivity (TFP) growth. Successful PSE reforms are expected to lift growth to 7 percent in the medium term driven by FDI generated exports.

- **Fiscal deficit.** The overall fiscal deficit projections are somewhat higher in the medium term, but in the long term they are in line with the authorities' commitment to meet the key WAEMU convergence criterion on the fiscal deficit (see paragraph 10 below)

- **Current account deficit**: The current account deficit is projected to narrow gradually from 10.3 percent of GDP in 2014 to just above 7.4 percent in 2019 and further down in the long term. This would be driven by projected fiscal consolidation and stronger dynamism in exports (mining in particular). Remittances are projected to remain significant as a share of GDP.

[3] The Senegalese authorities' new development strategy,

- **Inflation:** it is expected to remain moderate, on average less than 1.4 percent in the medium term.

Table 3. Evolution of Selected Macroeconomic Indicators

	2012	2013	2014	2015	Long term 1/
Real GDP growth					
Current DSA	3.4	3.5	4.5	4.6	7.3
Previous DSA	3.5	3.5	4.9	5.0	5.4
Overall fiscal deficit (percent of GDP)					
Current DSA	5.9	5.5	5.2	4.7	2.6
Previous DSA	5.9	5.5	5.1	3.9	2.6
Current account deficit (percent of GDP)					
Current DSA	10.8	10.9	10.3	8.8	7.5
Previous DSA	10.8	10.4	10.0	8.9	7.6

1/ Defined as the last 15 years of the projection period. For the current DSA update, the long term covers the years 2020-2034 (same as the full DSA in June 2014).

- **Financing:** The financing assumptions under this DSA are broadly similar to those under the most recent DSA (July 2014). As noted in the previous DSA, the authorities are increasingly relying on external nonconcessional or semi-concessional borrowing to finance infrastructure projects, and this trend is expected to continue. At the time of the previous DSA, the authorities had postponed a planned Eurobond issuance from 2013 to 2014 following a sharp tightening of financial conditions on international markets in the course of 2013. The authorities have since issued the US$500 million Eurobond in July 2014. Conditions have been relatively favorable in international markets in the past few months, and the authorities got a rate of 6.25 percent compared to the 6 percent yield on the 2011 Eurobond. This interest rate is higher than expected at the time of the previous DSA, partly owing to market concerns about the slow pace of reform. However, part of the proceeds would be used to repay the euro tranche of the syndicated loan contracted in 2013, which has a shorter maturity and higher rate (6.5 percent). The projections assume a repayment of the 2011 and 2014 bonds at maturity, as well as a moderate annual amount of non-concessional borrowing in the medium and long term. The authorities intend to continue relying on semi-concessional project financing (i.e. with a grant element above 15 percent) for infrastructure. As a result, the average grant element of new external borrowing is projected to decrease from about 20 percent to just above 10 percent over the projection period, as Senegal gradually moves away from concessional borrowing toward non-concessional borrowing.

- **Discount rate:** A discount rate of 5 percent has been used for this DSA.
- **Alternative Scenarios.** In addition to the baseline, scenario, the DSA considers alternative scenarios using the authorities' PSE projections (with higher debt accumulation in the early years.The probability approach is also applied.

EXTERNAL DSA

4. The use of remittances in the base case is justified for Senegal, since remittances are both stable and high as a percentage of GDP and exports.[4] Since 2000, remittances have grown every year with the exception of 2009, when they fell 6 percent. Over the period 2010–2013, remittances in Senegal averaged 52 percent of exports of goods and services and 13 percent of GDP. They have become an important and reliable source of foreign exchange in Senegal, a pattern that is expected to continue.[5]

5. Under the baseline scenario (Figure 1), and taking remittances into account, debt burden indicators remain well below their thresholds The external PPG debt ratios remain below their respective thresholds even under the most extreme stress tests, with one exception. Two spikes in debt service reflect the assumption of the repayment of the Eurobonds at maturity, and lead to one breach under the most extreme stress test (a 30 percent depreciation of the currency). Given that the largest breach falls within a 10-percent band of the threshold, the probability approach was also applied.

6. Alternatively, a more rapid scaling up of spending would imply larger fiscal deficits, higher debt accumulation, and some deterioration in debt burden indicators[6]. Although the indicators remain below their policy-dependent thresholds, the PV of debt to GDP ratio and the PV of debt to exports ratio come closer to the thresholds (Figure 2), despite higher assumed growth than in the baseline. This suggests that, in a scenario where the spending is scaled up quickly, but expected growth dividends do not materialize, Senegal could be at risk of losing its "low risk" rating. It also indicates a need for caution in the recourse to non-concessional borrowing.

7. The probability of debt distress also appears to be low (Figure 3). Under the probability approach, which focuses on the evolution of the probability of debt distress over time based on a country's individual CPIA score and average GDP growth rate, all the indicators for Senegal remain below the thresholds in all scenarios, supporting the case for a low risk of debt distress.

[4] In line with the Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries, remittances must be presented as the base case in the DSA if they are both greater than 10 percent of GDP and greater than 20 percent of exports of goods and services.

[5] Both ratios are measured on a backward-looking, three-year average basis.

[6] This scenario assumes that higher capital spending boosts the fiscal deficit by about 0.7 percentage point of GDP in 2015-19, growth is assumed to increase by 1 percentage point over the same period.

PUBLIC DSA

8. Under the PSI baseline scenario, indicators of overall public debt (external plus domestic) do not show significant vulnerabilities. The PV of total public debt to GDP and the PV of total public debt to revenue gradually decrease over the projection period. The PV of public debt to GDP remains well below the benchmark level of 74 percent associated with public debt vulnerabilities for strong performers. Similar to the thresholds for PPG external debt, the benchmarks for total public debt vary depending on a country's CPIA score and designate levels above which the risk of public debt distress is heightened. The benchmarks are in PV terms. Benchmarks for total public debt differ from thresholds for PPG external debt in that they serve as reference points for triggering a deeper discussion of public domestic debt. Thresholds for PPG external play a fundamental role in the determination of the external risk rating. The public debt benchmark for Senegal is higher than in the previous DSA of June 2014, owing to Senegal's CPIA reclassification as a strong performer. The authorities' effort to increase maturities (from slightly over one year at the time of the previous DSA) should reduce exposure to rollover and interest rate risks in the context of financing from the regional market.

9. The public debt outlook would be much less favorable in the absence of fiscal consolidation (Table 2b). In a scenario that assumes an unchanged primary deficit (as a percent of GDP) over the entire projection period, the PV of public debt to GDP grows but does not breach the 74 percent benchmark level. The benchmark level is breached in the "historical" scenario (holding real GDP growth and the primary deficit constant at their historical levels). While overall the risks remain low, these stress tests highlight the importance of continuing the fiscal effort and raising growth.

CONCLUSION

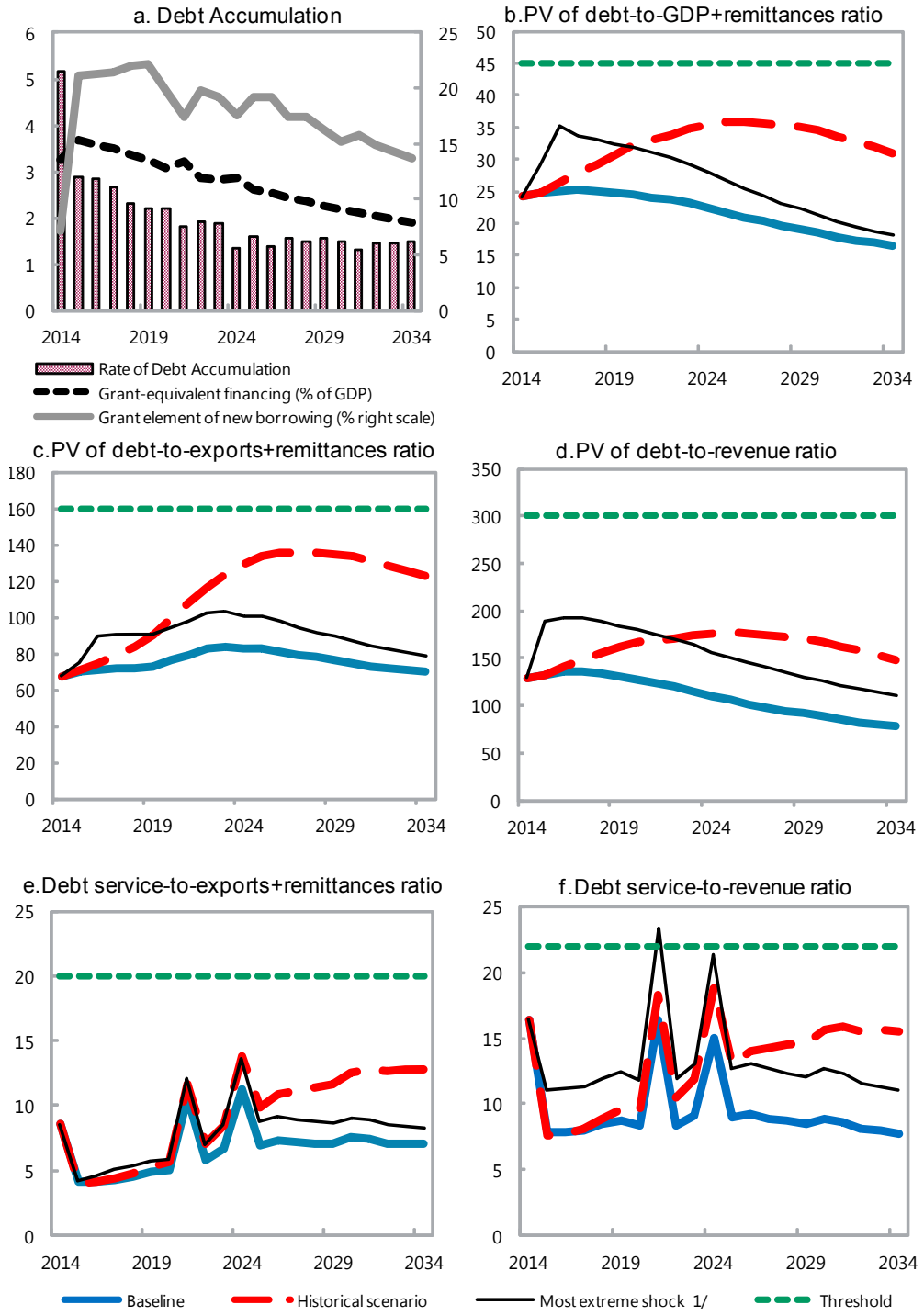
10. In staff's view, Senegal continues to face a low risk of debt distress. This assessment, however, hinges critically on a continued reduction of the fiscal deficit and prudence in the shift towards less concessional financing. Fiscal reforms should continue and additional fiscal space for PSE-related and social spending should be secured through efforts to increase revenue—particularly collecting tax arrears, freezing public consumption in real terms, and improving the composition of spending. The authorities also need to focus spending on productive areas, working closely with development partners to strengthen project design, preparation and execution while ensuring the overall quality and efficiency of public investment.

11. A cautious approach to non-concessional borrowing will similarly be essential for safeguarding debt sustainability. Preserving debt sustainability under the PSE as originally envisaged would depend on achieving a high growth dividend and implementing a comprehensive and ambitious reform of the state (to make room for investment and improve the efficiency of spending).

12. The conclusion also hinges on achieving projected growth, although there are some downside risks. The authorities are strongly committed to achieving **s**uccessful PSE reforms. These could lift growth to 7 percent in the medium term, driven by FDI generated exports. The PSE offers an achievable development strategy, including the right mix of private investment to be crowded in by public investment

in both human capital and infrastructure. However, unlocking private investment, including FDI, requires speeding up reforms to the business climate and improving public sector governance. Frontloading public investment without implementing the necessary structural reforms may jeopardize fiscal targets and debt sustainability while failing to raise growth from its sub-par trend. The main risks relate mainly to weak or slow implementation of the reforms, revenue shortfalls that would not allow sufficient mobilization of resources in support of the plan, failure to curb unproductive public consumption, and delays in raising expenditure efficiency, in particular of domestically financed capital expenditure.

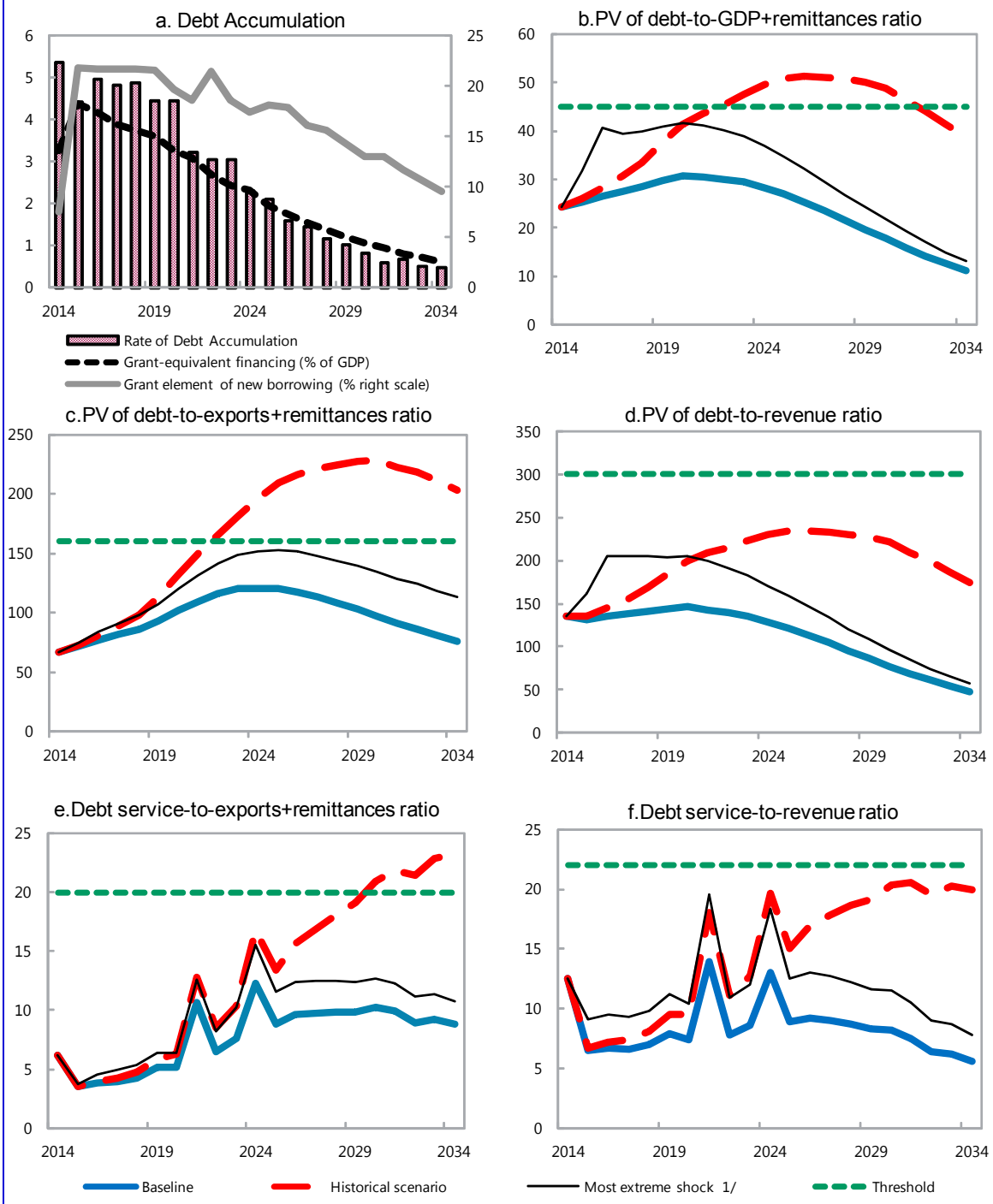
Figure 1. Senegal: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios 2014-2034 (Baseline with Remittances)



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

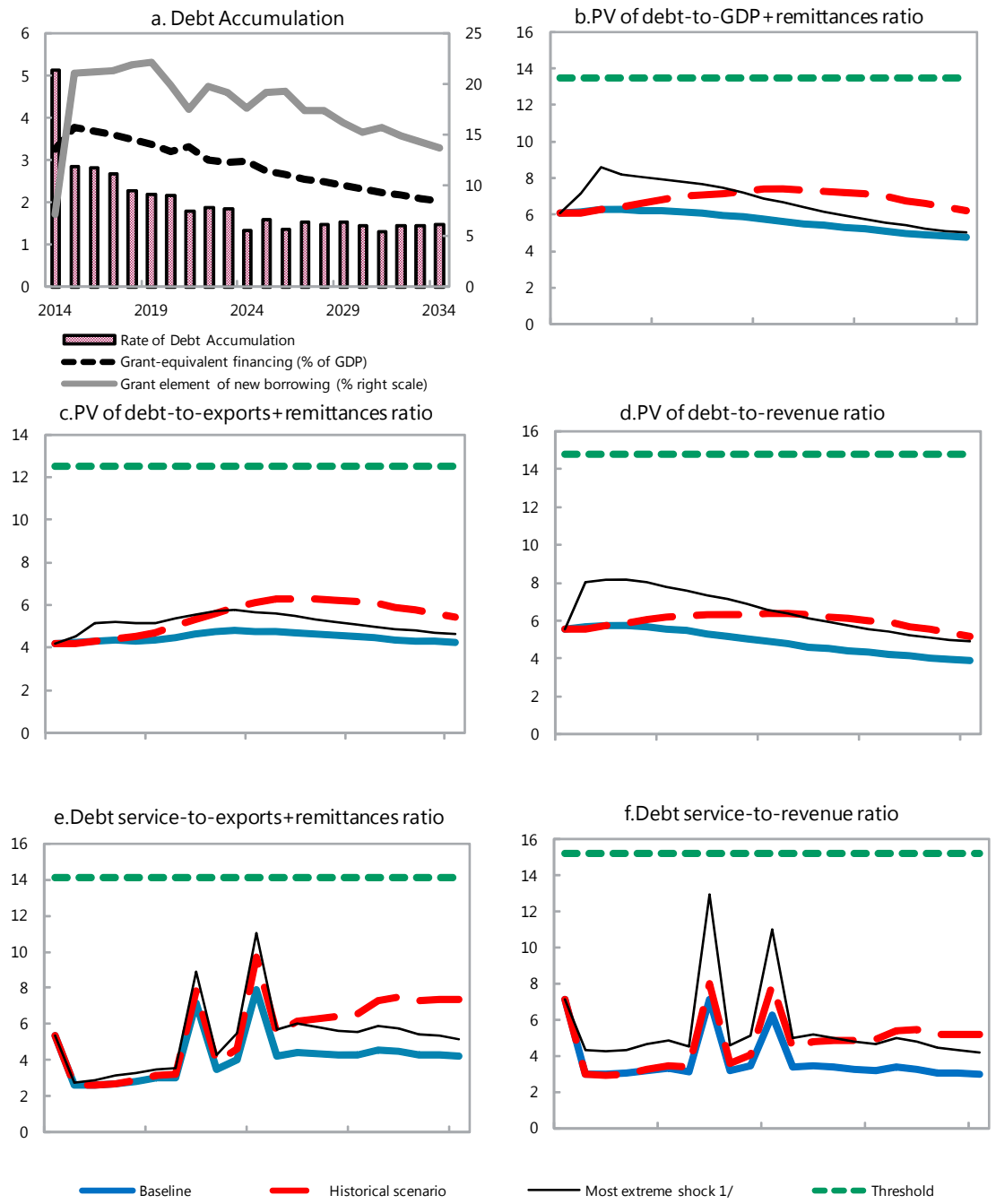
Figure 2. Senegal: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios (with higher capital spending), 2014-2034



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Combination shock; in c. to a Terms shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 3. Senegal: Probability of Debt Distress of Public and Publicly Guaranteed External Debt Under Alternative Scenarios 2014-2034



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Table 1. Senegal: External Debt Sustainability Framework, Baseline Scenario, 2011-2034

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**54.5**	**62.7**	**70.1**			**73.1**	**76.7**	**77.9**	**78.2**	**76.7**	**75.4**		**65.1**	**60.8**	
of which: public and publicly guaranteed (PPG)	29.4	31.0	32.7			35.0	35.8	36.4	36.5	36.0	35.3		30.1	21.2	
Change in external debt	3.8	8.1	7.4			3.0	3.6	1.2	0.3	-1.4	-1.3		-3.0	-0.2	
Identified net debt-creating flows	0.8	10.1	5.9			5.4	3.6	2.5	1.9	1.0	0.3		0.4	0.9	
Non-interest current account deficit	**6.9**	**9.9**	**10.0**	**8.3**	**2.9**	**9.3**	**7.8**	**7.2**	**7.0**	**6.6**	**6.4**		**6.3**	**6.7**	6.7
Deficit in balance of goods and services	18.3	21.3	21.2			20.5	19.6	19.1	19.0	18.5	18.1		14.0	12.9	
Exports	26.4	28.3	27.9			27.1	26.5	26.6	26.2	26.0	25.5		20.6	18.5	
Imports	44.7	49.5	49.1			47.6	46.1	45.7	45.2	44.5	43.6		34.6	31.4	
Net current transfers (negative = inflow)	-12.3	-12.5	-12.4	-10.9	2.1	-12.3	-12.8	-12.9	-12.8	-12.7	-12.4		-8.3	-6.5	-7.9
of which: official	-0.9	-1.0	-0.6			-0.8	-0.9	-0.9	-1.0	-1.0	-1.0		-0.8	-0.7	
Other current account flows (negative = net inflow)	0.9	1.2	1.3			1.1	1.0	0.9	0.8	0.8	0.7		0.5	0.3	
Net FDI (negative = inflow)	**-2.0**	**-2.0**	**-1.8**	**-1.8**	**0.6**	**-1.9**	**-2.0**	**-2.1**	**-2.1**	**-2.2**	**-2.3**		**-2.3**	**-2.3**	-2.3
Endogenous debt dynamics 2/	**-4.1**	**2.2**	**-2.3**			**-2.0**	**-2.2**	**-2.6**	**-3.0**	**-3.4**	**-3.9**		**-3.6**	**-3.5**	
Contribution from nominal interest rate	1.0	0.9	0.9			1.0	1.0	1.0	1.0	1.0	1.0		0.8	0.7	
Contribution from real GDP growth	-0.8	-1.9	-2.1			-3.0	-3.2	-3.6	-4.0	-4.4	-4.9		-4.5	-4.2	
Contribution from price and exchange rate changes	-4.3	3.2	-1.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**3.0**	**-2.0**	**1.5**			**-2.4**	**0.1**	**-1.3**	**-1.6**	**-2.4**	**-1.6**		**-3.4**	**-1.1**	
of which: exceptional financing	-0.9	-1.4	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	60.5			65.5	69.0	70.1	70.4	69.1	68.1		59.2	57.2	
In percent of exports	...	...	216.5			241.5	260.1	263.9	269.1	266.2	267.3		287.8	308.9	
PV of PPG external debt	**...**	**...**	**23.1**			**27.4**	**28.1**	**28.6**	**28.7**	**28.4**	**28.0**		**24.3**	**17.6**	
In percent of exports	**...**	**...**	**82.8**			**101.0**	**105.9**	**107.6**	**109.8**	**109.3**	**110.0**		**117.8**	**95.2**	
In percent of government revenues	**...**	**...**	**114.8**			**129.7**	**132.2**	**135.5**	**135.5**	**133.2**	**129.9**		**110.2**	**77.6**	
Debt service-to-exports ratio (in percent)	**17.8**	**9.7**	**10.3**			**18.4**	**7.9**	**8.2**	**8.0**	**8.6**	**8.1**		**16.3**	**8.2**	
PPG debt service-to-exports ratio (in percent)	**13.2**	**7.0**	**6.6**			**12.8**	**6.3**	**6.2**	**6.4**	**6.9**	**7.4**		**16.1**	**9.6**	
PPG debt service-to-revenue ratio (in percent)	**17.2**	**9.6**	**9.2**			**16.4**	**7.8**	**7.8**	**8.0**	**8.4**	**8.8**		**15.0**	**7.8**	
Total gross financing need (Billions of U.S. dollars)	1.4	1.5	1.6			2.0	1.3	1.3	1.4	1.4	1.5		2.8	6.2	
Non-interest current account deficit that stabilizes debt ratio	3.1	1.8	2.6			6.3	4.2	6.0	6.7	8.0	7.8		9.3	6.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.7	3.4	3.5	3.8	1.4	4.5	4.6	5.1	5.6	6.2	7.0	5.5	7.2	7.6	7.3
GDP deflator in US dollar terms (change in percent)	9.3	-5.5	1.8	4.3	8.0	2.7	1.4	2.8	3.1	3.0	2.3	2.6	2.7	3.1	2.8
Effective interest rate (percent) 5/	2.2	1.6	1.5	1.8	0.4	1.5	1.5	1.4	1.4	1.4	1.4	1.4	1.4	1.3	1.3
Growth of exports of G&S (US dollar terms, in percent)	17.7	4.7	4.1	9.0	10.6	4.2	3.8	8.2	7.2	8.4	7.5	6.6	9.1	9.9	8.1
Growth of imports of G&S (US dollar terms, in percent)	23.2	8.2	4.4	11.9	17.2	4.1	2.8	7.1	7.6	7.5	7.3	6.1	6.5	9.9	8.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	7.1	21.1	21.2	21.4	22.0	22.1	19.2	17.6	13.6	17.1
Government revenues (excluding grants, in percent of GDP)	20.3	20.4	20.1			21.1	21.2	21.1	21.2	21.3	21.6		22.0	22.7	22.2
Aid flows (in Billions of US dollars) 7/	0.7	1.0	0.8			0.5	0.5	0.6	0.6	0.6	0.7		0.9	1.9	
of which: Grants	0.3	0.4	0.4			0.5	0.5	0.5	0.5	0.6	0.6		0.8	1.6	
of which: Concessional loans	0.4	0.6	0.4			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.2	3.7	3.6	3.5	3.4	3.3		2.9	1.9	2.5
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			38.2	52.7	53.4	53.8	55.6	56.0		45.9	45.2	49.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	14.4	14.0	14.8			15.9	16.9	18.2	19.8	21.7	23.7		38.0	103.9	
Nominal dollar GDP growth	11.1	-2.3	5.4			7.3	6.1	8.0	8.9	9.3	9.5	8.2	10.1	11.0	10.3
PV of PPG external debt (in Billions of US dollars)			3.5			4.3	4.7	5.2	5.7	6.2	6.6		9.2	18.3	
(PVt-PVt-1)/GDPt-1 (in percent)						**5.1**	2.9	2.8	2.7	2.3	2.2	3.0	1.3	1.5	1.6
Gross workers' remittances (Billions of US dollars)	1.9	1.9	2.0			2.1	2.3	2.5	2.7	2.9	3.1		3.3	7.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	20.3			24.1	24.7	25.1	25.3	25.0	24.7		22.3	16.5	
PV of PPG external debt (in percent of exports + remittances)	...	...	55.5			67.6	69.7	70.8	72.1	71.9	72.4		83.0	69.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	4.4			8.6	4.1	4.1	4.2	4.5	4.9		11.3	7.0	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2014-2034

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	27	28	29	29	28	28	**24**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	27	28	30	32	33	35	**39**	34
A2. New public sector loans on less favorable terms in 2014-2034 2	27	29	30	31	31	31	**30**	26
Authorities' PSE framework	26	31	33	36	38	41	**33**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	27	29	30	30	30	29	**25**	18
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	27	29	33	33	32	32	**27**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	27	30	32	32	32	31	**27**	20
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	27	32	37	37	36	35	**29**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	27	32	39	39	38	37	**31**	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	27	40	41	41	40	40	**34**	25
PV of debt-to-exports ratio								
Baseline	101	106	108	110	109	110	**118**	95
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	101	106	112	121	127	137	**188**	182
A2. New public sector loans on less favorable terms in 2014-2034 2	101	109	114	118	120	123	**145**	138
Authorities' PSE framework	95	115	124	135	143	151	**137**	82
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	101	106	108	110	110	110	**118**	95
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	101	117	144	146	144	144	**150**	112
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	101	106	108	110	110	110	**118**	95
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	101	121	139	140	138	138	**140**	98
B5. Combination of B1-B4 using one-half standard deviation shocks	101	118	143	145	143	143	**145**	102
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	101	106	108	110	110	110	**118**	95
PV of debt-to-revenue ratio								
Baseline	130	132	135	136	133	130	**110**	78
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	130	133	141	149	155	161	**176**	149
A2. New public sector loans on less favorable terms in 2014-2034 2	130	136	143	146	146	145	**136**	113
Authorities' PSE framework	134	156	166	177	189	203	**164**	107
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	130	135	143	143	140	136	**116**	81
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	130	138	156	155	152	147	**121**	79
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	130	139	153	153	150	146	**124**	87
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	130	152	174	173	169	163	**131**	80
B5. Combination of B1-B4 using one-half standard deviation shocks	130	151	185	183	179	173	**139**	86
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	130	188	193	193	190	184	**157**	110

Table 2. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2014-2034 (concluded)

Debt service-to-exports ratio								
Baseline	13	6	6	6	7	7	**16**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	13	6	6	7	7	8	**20**	19
A2. New public sector loans on less favorable terms in 2014-2034 2/	13	6	6	6	6	7	**11**	12
Authorities' PSE framework	9	7	8	8	9	9	**10**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	13	6	6	6	7	7	**16**	10
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	13	7	7	8	9	9	**20**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	13	6	6	6	7	7	**16**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	13	6	7	8	8	8	**19**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	13	6	7	8	8	9	**20**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	13	6	6	6	7	7	**16**	10
Debt service-to-revenue ratio								
Baseline	16	8	8	8	8	9	**15**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	16	8	8	8	9	10	**19**	16
A2. New public sector loans on less favorable terms in 2014-2034 2/	16	8	7	7	8	8	**10**	10
Authorities' PSE framework	12	9	11	11	11	12	**13**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	16	8	8	8	9	9	**16**	8
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	16	8	8	9	9	9	**16**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	16	8	9	9	9	10	**17**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	16	8	9	9	10	10	**18**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	16	8	9	10	10	11	**19**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	16	11	11	11	12	12	**21**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	16	16	16	16	16	16	**16**	16

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assumingan offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Senegal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011-2034

	Actual					Estimate						Projections			
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	40.7	43.4	47.1			49.7	51.0	51.7	52.0	51.8	50.8		41.9	24.9	
of which: foreign-currency denominated	29.4	31.0	32.7			35.0	35.8	36.4	36.5	36.0	35.3		30.1	21.2	
Change in public sector debt	5.2	2.7	3.7			2.6	1.3	0.7	0.3	-0.2	-1.0		-2.0	-1.3	
Identified debt-creating flows	4.9	3.2	3.3			3.6	1.2	0.5	-0.1	-0.6	-1.4		-2.1	-1.3	
Primary deficit	5.2	4.4	3.9	3.9	1.1	3.5	2.9	2.5	2.2	2.0	1.2	2.4	0.5	0.3	0.5
Revenue and grants	22.5	23.3	22.7			24.0	24.0	23.8	23.9	23.9	24.1		24.2	24.2	
of which: grants	2.2	2.9	2.6			2.8	2.8	2.7	2.7	2.6	2.5		2.1	1.5	
Primary (noninterest) expenditure	27.7	27.7	26.6			27.5	26.9	26.3	26.1	25.9	25.3		24.6	24.6	
Automatic debt dynamics	-0.3	-0.6	-0.6			0.1	-1.7	-2.0	-2.3	-2.6	-2.6		-2.6	-1.7	
Contribution from interest rate/growth differential	-0.4	-0.7	0.7			-0.7	-1.4	-1.8	-2.0	-2.4	-2.7		-2.6	-1.7	
of which: contribution from average real interest rate	0.2	0.6	2.2			1.3	0.8	0.7	0.7	0.7	0.7		0.4	0.2	
of which: contribution from real GDP growth	-0.6	-1.3	-1.5			-2.0	-2.2	-2.5	-2.7	-3.0	-3.4		-3.0	-1.9	
Contribution from real exchange rate depreciation	0.1	0.1	-1.3			0.8	-0.2	-0.2	-0.3	-0.2	0.1		...	...	
Other identified debt-creating flows	0.0	-0.6	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	-0.6	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.3	-0.5	0.4			-1.0	0.1	0.3	0.4	0.4	0.4		0.1	0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	37.5			42.1	43.3	44.0	44.3	44.2	43.5		36.1	21.3	
of which: foreign-currency denominated	...	...	23.1			27.4	28.1	28.6	28.7	28.4	28.0		24.3	17.6	
of which: external	...	...	23.1			27.4	28.1	28.6	28.7	28.4	28.0		24.3	17.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	12.3	12.2	12.2			15.7	13.4	13.1	12.3	12.6	10.2		9.0	4.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	165.3			175.9	180.1	184.5	185.4	184.9	180.5		149.6	88.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	186.4			199.6	203.7	208.3	208.8	207.5	201.6		164.1	93.9	
of which: external 3/	...	...	114.8			129.7	132.2	135.5	135.5	133.2	129.9		110.2	77.6	
Debt service-to-revenue and grants ratio (in percent) 4/	19.9	16.4	17.0			24.2	28.1	33.2	31.3	34.6	28.2		29.7	13.4	
Debt service-to-revenue ratio (in percent) 4/	22.1	18.7	19.2			27.5	31.7	37.5	35.3	38.9	31.6		32.6	14.3	
Primary deficit that stabilizes the debt-to-GDP ratio	0.0	1.7	0.2			0.9	1.5	1.8	2.0	2.2	2.3		2.5	1.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.7	3.4	3.5	3.8	1.4	4.5	4.6	5.1	5.6	6.2	7.0	5.5	7.2	7.6	7.3
Average nominal interest rate on forex debt (in percent)	3.7	2.5	2.5	2.4	0.6	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	3.5	3.0
Average real interest rate on domestic debt (in percent)	3.7	4.8	8.1	3.2	3.7	4.8	4.2	3.3	3.4	3.2	3.3	3.7	2.9	2.6	2.7
Real exchange rate depreciation (in percent, + indicates depreciation	0.3	0.5	-4.3	-0.8	8.5	2.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.2	2.3	-1.4	2.4	2.8	0.7	2.3	2.3	2.3	2.3	2.3	2.0	2.7	3.1	2.8
Growth of real primary spending (deflated by GDP deflator, in percer	7.9	3.2	-0.4	1.1	2.6	7.9	2.4	2.8	4.8	5.1	4.8	4.6	6.6	7.6	7.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	7.1	21.1	21.2	21.4	22.0	22.1	19.2	17.6	13.6	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

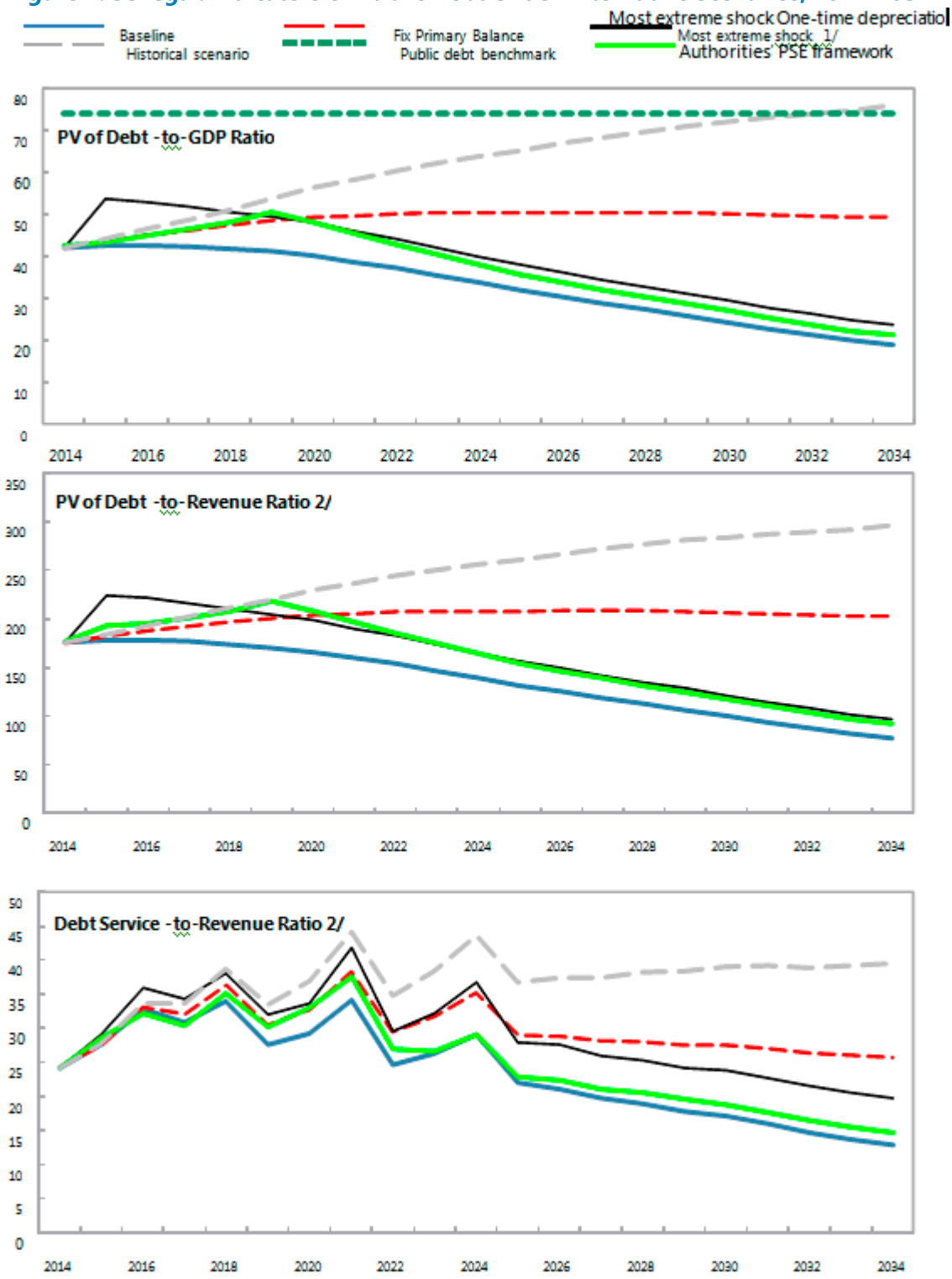
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Senegal: Sensitivity Analysis for Key Indicators of Public Debt, 2014-2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	42	43	44	44	44	43	36	21
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	42	45	47	49	52	55	65	77
A2. Primary balance is unchanged from 2014	42	44	45	47	48	49	51	50
A3. Permanently lower GDP growth 1/	42	43	44	45	45	45	40	31
Authorities' PSE framework	43	44	45	47	48	51	38	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	42	45	47	49	49	49	45	33
B2. Primary balance is at historical average minus one standard deviations in 2015-201	42	45	48	48	48	47	39	23
B3. Combination of B1-B2 using one half standard deviation shocks	42	45	48	49	50	49	44	31
B4. One-time 30 percent real depreciation in 2015	42	54	54	54	53	52	42	26
B5. 10 percent of GDP increase in other debt-creating flows in 2015	42	52	52	52	52	51	42	24
PV of Debt-to-Revenue Ratio 2/								
Baseline	176	180	184	185	185	180	150	88
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	176	185	196	206	215	224	261	304
A2. Primary balance is unchanged from 2014	176	183	191	196	201	204	212	207
A3. Permanently lower GDP growth 1/	176	181	186	188	189	186	164	128
Authorities' PSE framework	178	193	196	201	208	218	163	89
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	176	185	198	203	205	204	185	138
B2. Primary balance is at historical average minus one standard deviations in 2015-201	176	188	201	201	200	195	160	93
B3. Combination of B1-B2 using one half standard deviation shocks	176	188	203	206	207	204	180	126
B4. One-time 30 percent real depreciation in 2015	176	226	228	225	222	215	175	108
B5. 10 percent of GDP increase in other debt-creating flows in 2015	176	217	220	219	217	210	172	100
Debt Service-to-Revenue Ratio 2/								
Baseline	24	28	33	31	35	28	30	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	28	34	34	38	33	44	39
A2. Primary balance is unchanged from 2014	24	28	33	32	36	30	35	26
A3. Permanently lower GDP growth 1/	24	28	33	32	35	29	31	18
Authorities' PSE framework	24	29	32	30	35	30	29	14
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	24	29	35	33	37	31	34	19
B2. Primary balance is at historical average minus one standard deviations in 2015-201	24	28	34	34	38	29	31	14
B3. Combination of B1-B2 using one half standard deviation shocks	24	28	34	34	38	31	33	18
B4. One-time 30 percent real depreciation in 2015	24	30	36	35	39	33	38	20
B5. 10 percent of GDP increase in other debt-creating flows in 2015	24	28	35	42	37	32	32	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure 4. Senegal: Indicators of Public Debt Under Alternative Scenarios, 2014-2034



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.